

# Computer Vision for CS & AI

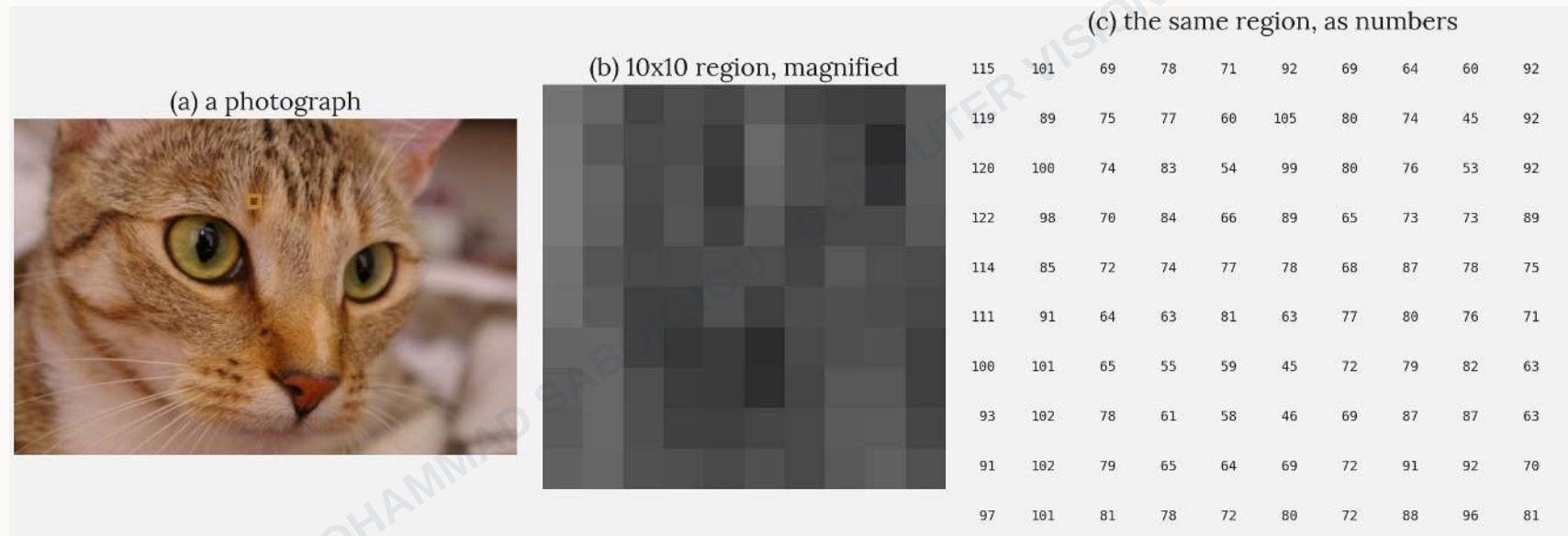
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## Session 1 — Images as Numbers

*Why a problem your brain solves in a seventh of a second took the field forty years*

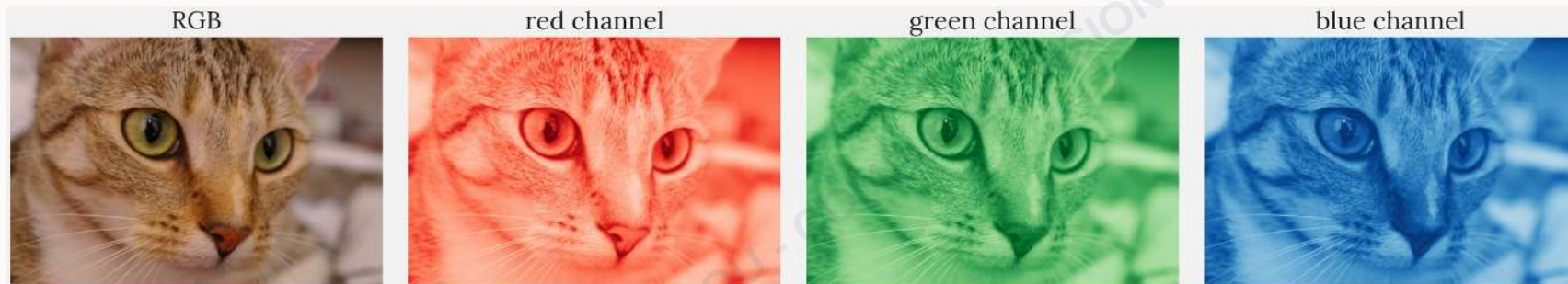
## THE QUESTION

# You knew before you decided anything



*A photograph, a 10×10 patch near the eye, and that patch printed as numbers. Panel (c) is the entire input a computer receives.*

# Colour is three grids, not one



## Grayscale

a function  $I(r,c)$  on a grid — shape (H, W)

## Colour

three stacked grids — shape (H, W, 3)

## 8-bit depth

every value an integer in [0, 255]

**405,900**

numbers in this one  
small photograph

# The idea every student has first

*“To recognise an object, store a picture of it and compare new images pixel by pixel.”*

It is a good proposal, because its failure is instructive.

1

**Store**

one template image per object

2

**Compare**

mean absolute difference, pixel by pixel

3

**Decide**

nearest template wins

WHY IT IS HARD

# Same cup of coffee, six times

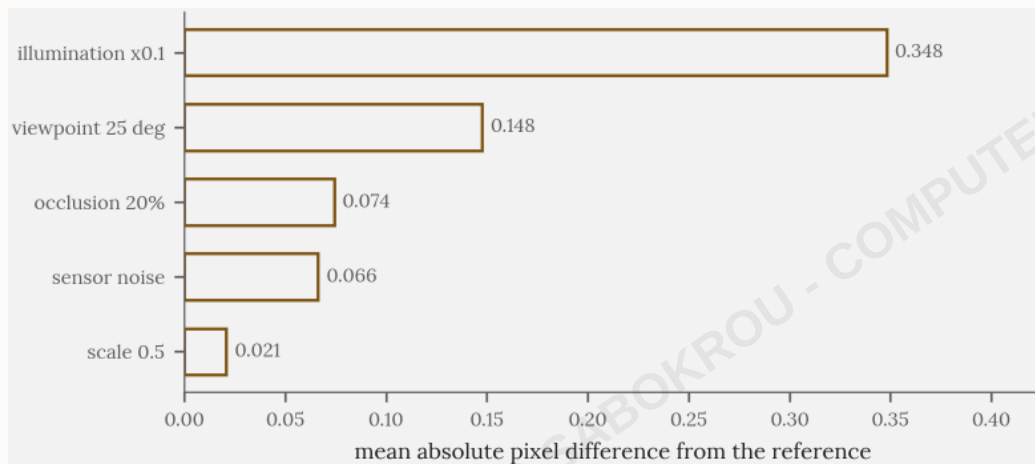


**Illumination • Viewpoint • Scale • Occlusion • Sensor noise • Deformation • Background clutter • Intra-class variation**

*You classified all six correctly without effort. Learn these eight by name — you will use them all semester to diagnose why a model is failing.*

## THE RESULT

# Now measure what the machine measures



Dimming the lights

changes the data more than

**covering a fifth  
of the cup**

A rule based on pixel distance calls a dark photo of the cup less like the cup than a heavily occluded one.

*In Lab 1 this same effect takes a working classifier from 100% to 10% accuracy — on images a human still reads perfectly.*

SO WHY NOT JUST STORE EVERY IMAGE?

$10^{362,000}$

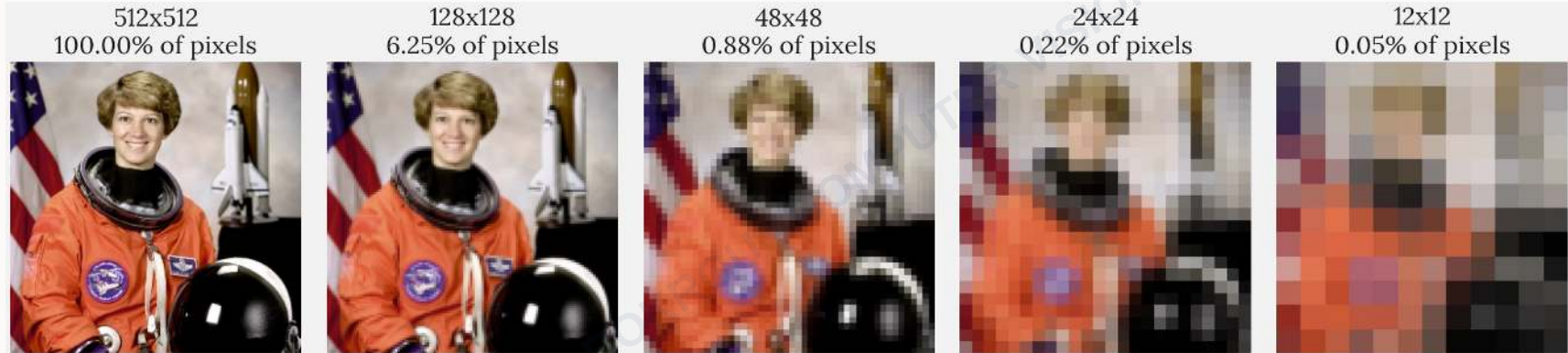
possible 224×224 colour images

$10^{80}$

atoms in the observable universe

*Memorising images is not inefficient. It is arithmetically impossible — so a working system must compute a representation instead.*

# How little is actually enough?

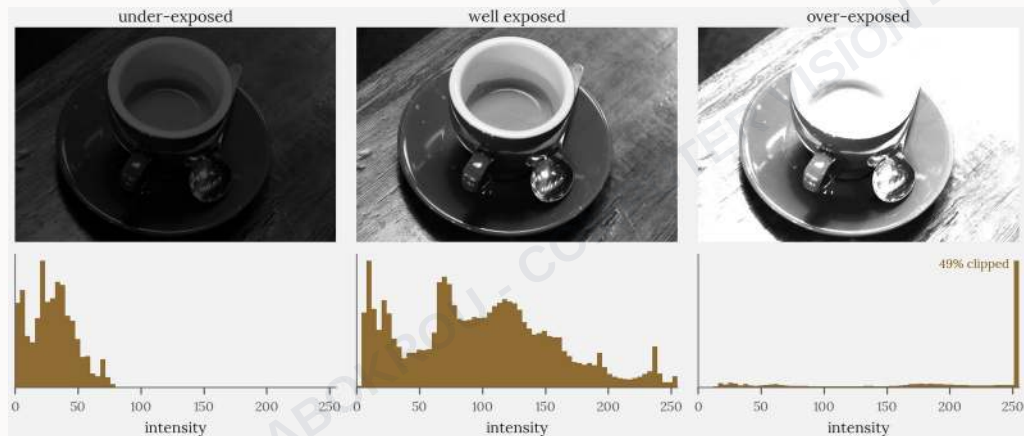


**576**

pixels — 0.22% of the original — and most people still see a person.

The information that identifies an object survives massive compression, so a compact representation is at least possible.

# The data is made, not found



*One scene, three exposures. The scene is identical; the data is not — the histograms barely overlap.*

A model trained only on well-exposed images has never seen the other two, and will behave unpredictably when it meets them.

WHEN THE HISTOGRAM IS A PERSON

## In Lab 1 you will measure this yourself

20%

reported overall

40%

group A

0%

group B

*Nobody wrote a rule excluding group B. The exclusion is a consequence of reporting an average.*

Buolamwini & Gebru (2018) found under 1% error for lighter-skinned men and up to ~35% for darker-skinned women, in systems whose reported overall accuracy looked good.

# Where this course goes

## CH 1-2

**Images, light  
and histograms**

What the data is

## CH 3-4

**Filtering, edges,  
features**

Hand-designed  
representations

## CH 5-9

**Learning, CNNs,  
depth**

Learned  
representations

## CH 10-15

**Detection,  
segmentation,  
tracking, evaluation**

Systems, and  
judging them

Every week: two hours of lecture, two hours of lab. The lab is where the ideas become yours.

# Before we meet again

0

## Lab 0 — Getting Set Up

Two hours. Do this **BEFORE** Lab 1.

- Colab running, saved to your Drive
- A dataset loaded and inspected
- One image read, one image written
- Results saved and read back
- A green environment report

## Read

Chapter 1, including the worked examples. They are short and they are the lecture in slow motion.

## Bring

A laptop that can open a browser. Nothing is installed locally — all work happens in Colab.

*If Lab 0 does not end with a green report, come to the first ten minutes of Lab 1 and we will fix it there.*

Next week

**You will build the classifier from slide 4,  
watch it score 100%, and then break it  
with a light dimmer.**

*Then you will repair it with one line — and that line is the first representation you will ever write.*